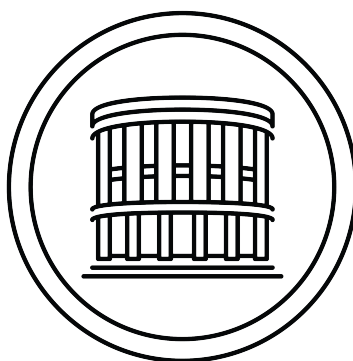


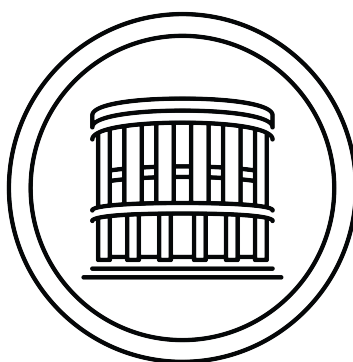
COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



CONSTRUCTIONS OF HAMILTONIAN GRAPHS WITH PRESCRIBED PARAMETERS

Master thesis

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FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



CONSTRUCTIONS OF HAMILTONIAN GRAPHS WITH PRESCRIBED PARAMETERS

Master thesis

Study program: Applied Informatics
Branch of study: Applied Informatics
Department: Department of Applied Informatics
Supervisor: doc. RNDr. Tatiana Jajcayová, PhD



THESIS ASSIGNMENT

Name and Surname: Bc. Jakub Šmihul'a
Study programme: Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study: Computer Science
Type of Thesis: Diploma Thesis
Language of Thesis: English
Secondary language: Slovak

Title: Constructions of Hamiltonian graphs with prescribed parameters

Annotation: Constructions of cages and record graphs, as well as study of their properties, is a very active area of research in extremal graph theory. For each pair of parameters (k, g) there exists infinitely many (k, g) -graphs. The graph of the smallest order, $n(k, g)$, among these graphs is called a (k, g) -cage. Cages are known only for few pairs of parameters. It is computationally very difficult to prove that some (k, g) -graph is a cage. The smallest known (k, g) -graph, that was not yet proven to be a cage is called a record $\text{rec}(k, g)$ -graph. Sachs in his work conjectured that for each pair of parameters k, g , there exists a cage that is Hamiltonian. Therefore, investigating the Hamiltonicity of cages and record graphs is interesting.

Aim: The aim of the theses is to develop graph constructions that for any given (k, g) -graph will return hamiltonian (k, g) -graph. We will be interested in such graphs of smallest possible order, or at least in some upper bounds for orders of such graphs. This will require to create and program clever algorithms for graph constructions, and also run highly non-trivial computer experiments to test graph properties.

Keywords: extremal graphs, girth, valency, hamiltonicity

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Assigned: 19.11.2025

Approved: 19.11.2025
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ZADANIE ZÁVEREČNEJ PRÁCE

Meno a priezvisko študenta: Bc. Jakub Šmihul'a
Študijný program: aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor: informatika
Typ záverečnej práce: diplomová
Jazyk záverečnej práce: anglický
Sekundárny jazyk: slovenský

Názov: Constructions of Hamiltonian graphs with prescribed parameters
Konštrukcie hamiltonovských grafov s predpísanými parametrami

Anotácia: Konštrukcie kliebok a rekordných grafov, ako aj štúdium ich vlastností, sú veľmi aktívnou oblasťou výskumu v extrémálnej teórii grafov. Pre každú dvojicu parametrov (k, g) existuje nekonečne veľa (k, g) -grafov. Graf najmenšieho rádu, $n(k, g)$, spomedzi týchto grafov sa nazýva (k, g) -klietka. Klietky sú známe len pre niektoré parametre. Výpočtovo je veľmi ťažké dokázať, že nejaký (k, g) -graf je kliebkou. Najmenší známy (k, g) -graf, u ktorého sa ešte nedokázalo, že je kliebkou, sa nazýva rekordný $\text{rec}(k, g)$ -graf. Sachs vo svojej práci predpokladal, že pre každý pár parametrov k, g existuje klietka, ktorá je hamiltonovská. Preto je skúmanie hamiltonovskosti kliebok a rekordných grafov zaujímavé.

Cieľ: Cieľom diplomovej práce je navrhnúť konštrukcie grafov, ktoré pre akýkoľvek daný (k, g) -graf vrátia hamiltonovský (k, g) -graf. Budeme sa zaujímať o takéto grafy najmenšieho možného rádu, alebo aspoň o niektoré horné ohraničenia pre rády takýchto grafov. To si bude vyžadovať vytvorenie a naprogramovanie komplexných algoritmov pre konštrukcie grafov a tiež testovanie vlastností grafov pomocou netriviálnych počítačových experimentov.

Kľúčové slová: extrémálne grafy, obvod, stupeň, hamiltonicita

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Dátum zadania: 19.11.2025

Dátum schválenia: 19.11.2025

prof. RNDr. Roman Ďurikovič, PhD.
garant študijného programu

.....
študent

.....
vedúci práce

I hereby declare that I have written this thesis by myself, only with help of referenced literature, under the careful supervision of my thesis advisor.

Bratislava, 2026

.....
Bc. Jakub Smihula

Acknowledgement

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Abstract

[TODO: write English abstract.]

Keywords: TODO

Abstrakt

[TODO: napísať slovenský abstrakt.] ...

Kľúčové slová: TODO

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Chapter 1

Introduction

[TODO: write introduction.]

Chapter 2

Motivation

[TODO: write motivation.]

Chapter 3

Preliminaries and Definitions

[TODO: write Preliminaries and Definitions]

Chapter 4

Hamilton Cycles

[TODO: write Hamilton Cycles.]

Chapter 5

Constructions with Prescribed Degree Conditions

[TODO: write Constructions with Prescribed Degree Conditions.]

Chapter 6

Hardware and Software Tools

[TODO: write Hardware and Software Tools.]

Chapter 7

General Construction Techniques

[TODO: write General Construction Techniques.]

Chapter 8

New Constructions and Main Results

[TODO: write New Constructions and Main Results.]

Chapter 9

Conclusion

[TODO: write Conclusion.]

Bibliography

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